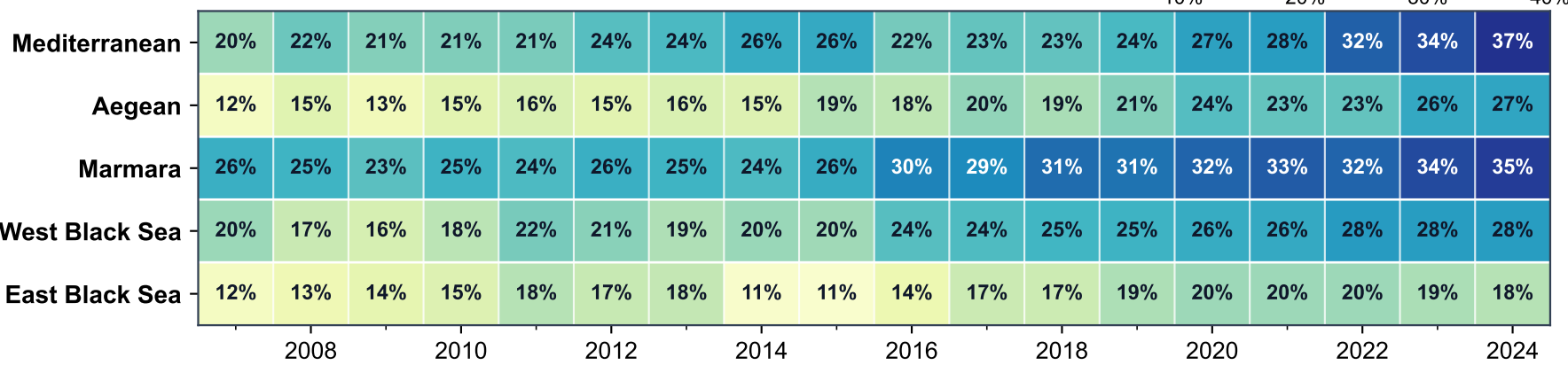
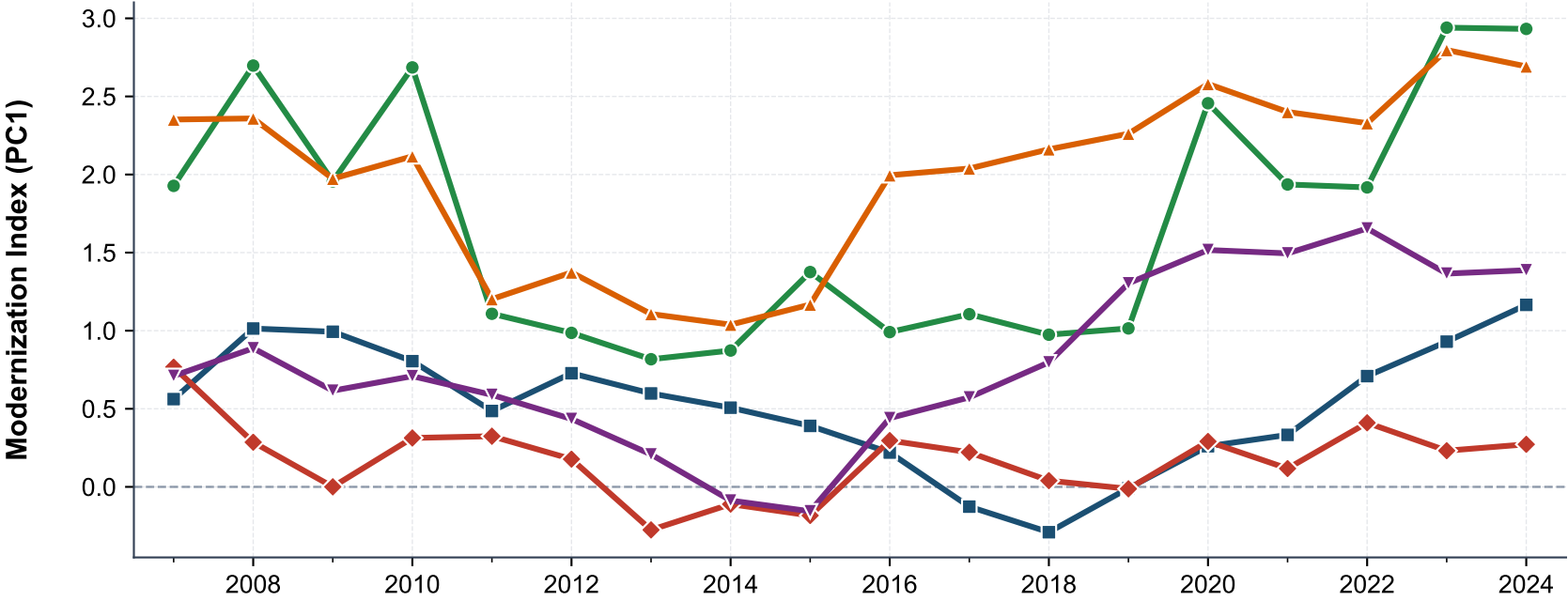


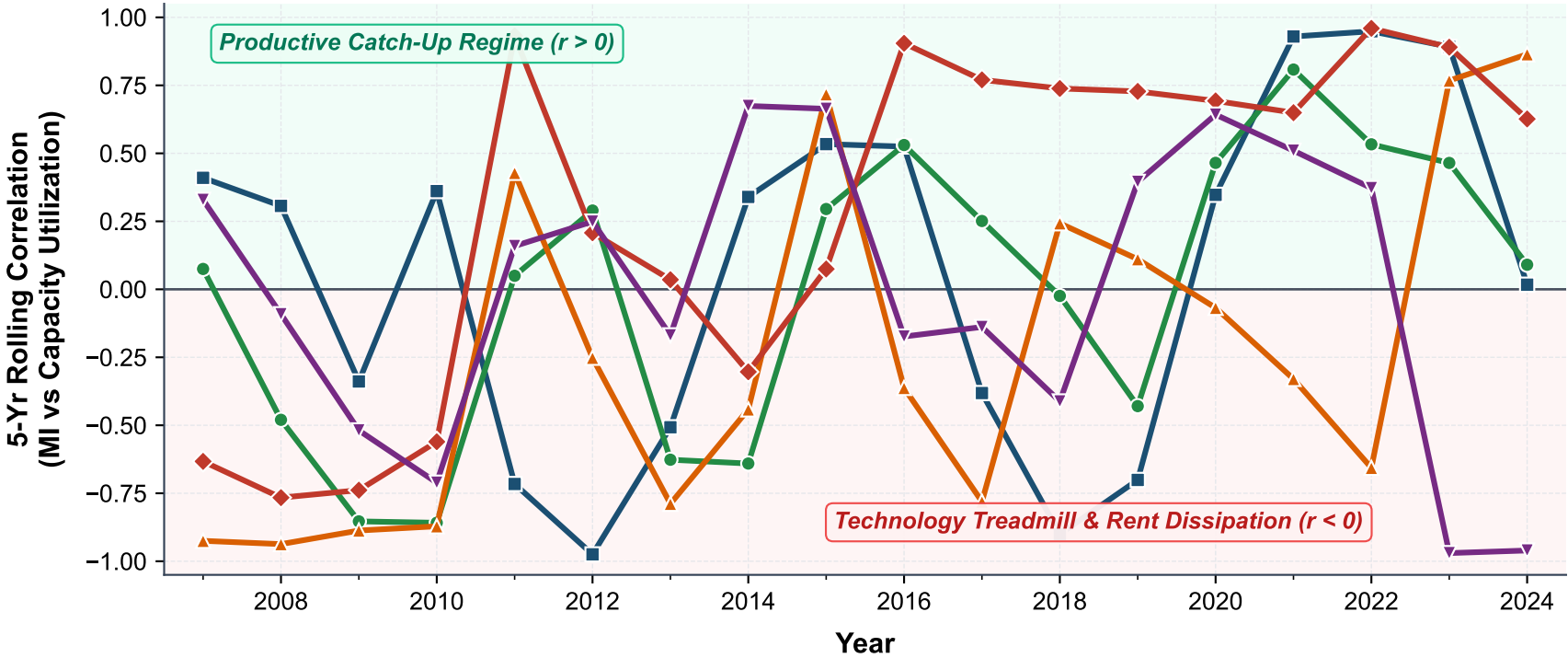
(a) Equipment Adoption Rate by Basin (2007–2024)



(b) Fleet Modernization Index (PC1) Trajectory (2007–2024)



(c) Dynamic Modernization–Efficiency Correlation (5-Year Rolling Window)



■ Mediterranean (Türkiye) ● Aegean (Türkiye) ▲ Marmara (Türkiye) ◆ West Black Sea (Türkiye) ▼ East Black Sea (Türkiye)